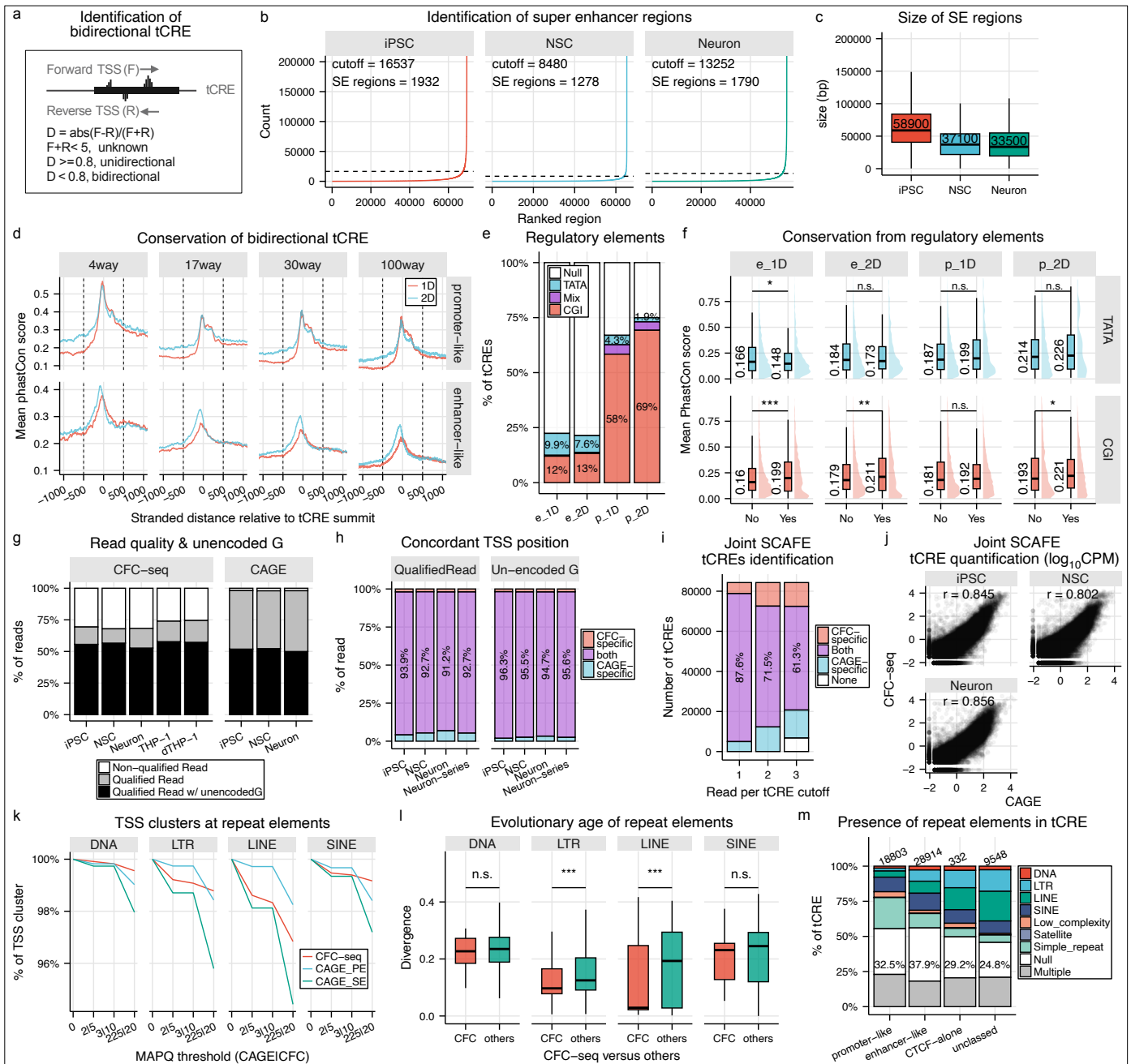




**Extended Data Figure 3 | Features of tCREs related to directionality, super enhancers and repetitive elements.**

**a**, Schematic illustrating the classification of tCREs as bidirectional (2D) or unidirectional (1D) based on transcription activity from both strands. **b**, Read count distribution of stitched H3K27ac peaks generated by ROSE. Regions exceeding the dotted line threshold were classified as super enhancers. **c**, Size distribution of super enhancers identified in the three analyzed cell types. **d**, Sequence conservation around tCRE summits for 2D and 1D promoters and enhancers, based on phastCons scores as described in **Extended Data Figure 2e**. **e**, Proportion of 2D and 1D tCREs (major strand only) containing CpG islands (CGI), TATA boxes, or both (Mix). **f**, Sequence conservation (phastCons 17-way scores) around the summit ( $\pm 500$  nt) of 2D and 1D, promoter-like and enhancer-like tCREs, stratified by presence of CGI or TATA box. Only major strand tCREs were included. **g**, Classification of mapped reads analyzed by SCAFE from CFC-seq and CAGE datasets into: not qualified (5' soft-clipping  $> 3$  nt), qualified without unencoded G, and qualified with unencoded G. **h**, Distribution of single-nucleotide TSSs found uniquely in CFC-seq, in both CFC-seq and CAGE, or uniquely in CAGE. **i**, Number of tCREs detected by CFC-seq, CAGE, or both at varying read count thresholds. tCREs were identified by joint SCAFE analysis of Neuron series CFC-seq and CAGE datasets. **j**, Expression correlation per cell type between CFC-seq and CAGE, shown as RLE-normalized CPM values for each tCRE (each dot represents one tCRE). tCREs were identified by joint SCAFE analysis. **k**, TSS cluster identification coverage by CFC-seq, CAGE paired-end (PE), and CAGE single-end (SE) across increasing MAPQ thresholds. Coverage is shown relative to union TSS clusters at MAPQ = 0. **l**, Repetitive element divergence (as a proxy for evolutionary age) for repeats overlapping CFC-specific TSS clusters at MAPQ 20 versus others. **m**, Proportion of major strand tCREs containing repetitive elements, colored by repeat class. "Multiple" indicates overlap with more than one repeat class. Unless specified, statistical tests were two-sided Wilcoxon;  $p < 0.05$  (\*),  $p < 0.01$  (\*\*),  $p < 0.001$  (\*\*\*), n.s. = not significant.